Test Series Code

06

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | DROPPER | **M. MARKS: 720**

DATE: 22/02/2026

SYLLABUS

Physics:	Electrostatic Potential and Capacitance, Current Electricity + 30% Questions from cumulative syllabus of previously conducted all AITS test
Chemistry:	Chemical Kinetics, Haloalkanes and Haloarenes, Alcohols, Ethers and Phenols + 30% Questions from cumulative syllabus of previously conducted all AITS test
Botany:	Sexual Reproduction in Flowering Plant, Molecular Basis of Inheritance + 30% Questions from cumulative syllabus of previously conducted all AITS test
Zoology:	Reproductive Health, Biomolecules, Human Health and Disease + 30% Questions from cumulative syllabus of previously conducted all AITS test

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. What is the dimensional formula of Electric Polarization (P)?

(1) $[M^0L^{-2}T^1A^1]$ (2) $[M^0L^{-3}T^1A^1]$

(3) $[M^1L^1T^{-3}A^{-1}]$ (4) $[M^0L^1T^1A^1]$

2. Two concentric thin spherical metallic shells have radii R and $2R$. The inner shell is given a charge $+Q$ and the outer shell is earthed. What is the charge induced on the inner surface of the outer shell?

(1) $+Q$ (2) $-Q$

(3) $\frac{-Q}{2}$ (4) Zero

3. A heater coil is rated 100 W, 200 V. It is cut into two identical parts. Both parts are connected in parallel across the same 200 V source. The new power consumed by the combination is:

(1) 100 W (2) 200 W

(3) 400 W (4) 25 W

4. A charged conductor has a surface charge density $\sigma = -17.7\times10^{-12}$ C/m^2. If the outward normal vector to the surface at a specific point just outside the surface of conductor is along the positive x-axis ($\hat{i}$), then what is the electric field vector at that point? (Take $\varepsilon_0 = 8.85\times10^{-12}$ C^2 / N$\cdot$m^2)

(1) $2\hat{i}$ N/C (2) $-2\hat{i}$ N/C

(3) $-2\hat{j}$ N/C (4) $0.5\hat{i}$ N/C

5. An electric dipole of dipole moment $\vec{p}$ is placed in a uniform electric field $\vec{E}$ such that $\vec{p}$ makes an angle of 60° with $\vec{E}$. The work done in rotating the dipole to a position where $\vec{p}$ is opposite to $\vec{E}$ is W. What is the magnitude of the torque acting on the dipole in its initial position?

(1) $\frac{W}{3}$ (2) $\frac{W}{\sqrt{3}}$

(3) $\sqrt{3}W$ (4) $\frac{2W}{3}$

6. A 3 m long ladder of mass 20 kg leans on a frictionless wall. Its feet rests on the rough floor at a distance 1 m from the wall. The normal forces of the wall and the floor respectively will be nearly: (Use g = 10 m/s^2)

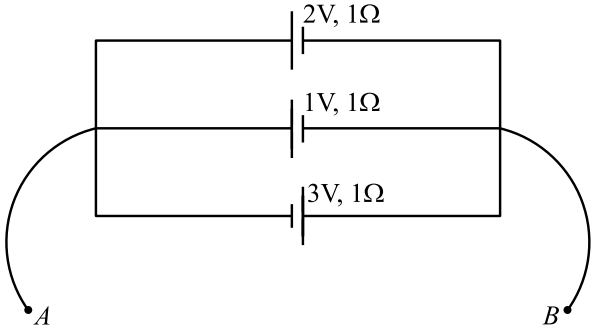
(1) 35 N and 200 N

(2) 30 N and 180 N

(3) 25 N and 200 N

(4) 45 N and 160 N

7. Calculate the equivalent emf of three different cells connected in parallel as shown below.



(1) 1 V (2) 2 V

(3) 3 V (4) 0 V

8. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): When two conductors of different shapes and sizes charged to different potentials are connected by a wire, energy is always lost.

Reason (R): The charge flows from higher potential to lower potential until potentials equalize, and during this flow, energy is dissipated as heat in the connecting wire.

In the light of the above statements, choose the **correct** answer from the options given below:

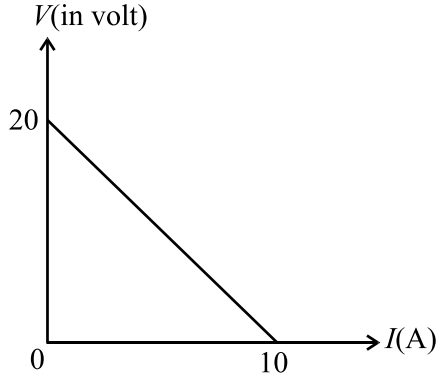
(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

9. A cell of emf E and internal resistance r is connected to a variable external resistance R. The figure below shows the variation of terminal voltage (V) of the cell with the current (I) in the circuit. What is the maximum power that can be delivered to the external resistance?



(1) 25 W (2) 50 W

(3) 100 W (4) 200 W

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10. Three identical charges $+q$ are placed at the vertices of an equilateral triangle of side L. How much work is required to move a fourth charge $+q$ from infinity to the centroid of the triangle?

(1) $\frac{3kq^2}{L}$ (2) $\frac{3\sqrt{3}kq^2}{L}$

(3) $\frac{\sqrt{3}kq^2}{L}$ (4) Zero

11. If the rate of emission of radiation from a hot ideal blackbody increases 256 times, the wavelength λ corresponding to maximum intensity of radiation becomes:

(1) $\frac{7}{16}$ of the initial value

(2) $\frac{1}{3}$ of the initial value

(3) $\frac{1}{4}$ of the initial value

(4) $\frac{1}{256}$ of the initial value

12. Two bulbs rated $(25\text{W}, 220\text{V})$ and $(100\text{W}, 220\text{V})$ are connected in series to a 440V supply. Which bulb will fuse?

(1) 25 W bulb

(2) 100 W bulb

(3) Both will fuse

(4) Neither will fuse

13. A galvanometer of resistance 100Ω gives full scale deflection for 10mA. To convert it into a voltmeter of range $(0-10)\text{V}$, the resistance to be connected in series is:

(1) 900Ω (2) 1000Ω

(3) 1100Ω (4) 9900Ω

14. A parallel plate capacitor with plate area A and separation d is charged to charge Q. The force of attraction between the plates is F. If the separation is doubled while keeping the charge constant, the new force will be:

(1) F (2) $2F$

(3) $\frac{F}{2}$ (4) $\frac{F}{4}$

15. A uniformly charged non-conducting solid sphere of radius R has an electric potential of 80 V at its surface. What is the electric potential at a point located at a distance $r = \frac{R}{2}$ from the center of the sphere? (Assume potential at infinity is zero).

(1) 120 V (2) 40 V

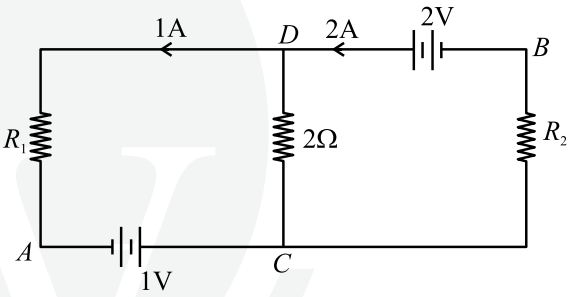
(3) 110 V (4) 100 V

16. A conducting sphere of radius 10cm has an unknown charge. If the electric field at a distance 20cm from the center of the sphere is $1.5\times10^3\ \text{N/C}$ and points radially inward, what is the net charge on the sphere?

(1) -6.67nC (2) $+6.67\text{nC}$

(3) -3.33nC (4) $+3.33\text{nC}$

17. Consider the circuit diagram shown below. The currents in the respective branches are given. If the electric potential at point A is assumed to be zero, what is the electric potential at point B (V_B)?



(1) 2 V (2) –3 V

(3) 1 V (4) 3 V

18. A uniform electric field exists in a region and is given by $\vec{E} = \left(3\hat{i} + 4\hat{j}\right)\text{N/C}$. Two points A and B are located at coordinates $(0\text{m}, 2\text{m})$ and $(2\text{m}, 0\text{m})$ respectively. Calculate the potential difference $\left(V_A - V_B\right)$.

(1) -2V (2) $+2\text{V}$

(3) -14V (4) $+14\text{V}$

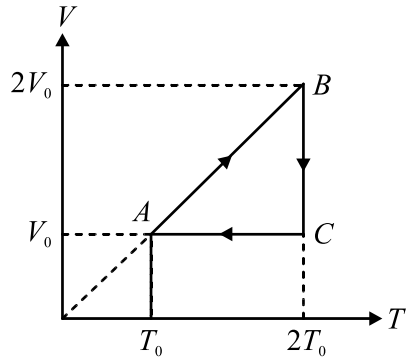
19. A positive point charge q of mass m is released from rest at a distance r from a fixed positive point charge Q. What will be the Kinetic Energy (K.E.) of charge q when it reaches infinity? (Neglect gravitational forces)

(1) $\frac{kQm}{qr}$ (2) $\frac{kQq}{r}$

(3) $\frac{kQq}{r^2}$ (4) $\frac{kQq}{mr}$

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20. An ideal monoatomic gas undergoes a cyclic process *ABCA* as shown in the figure. The ratio of magnitude of heat absorbed during process *AB* to the magnitude of work done on the gas during *BC* is:



(1) $\frac{5}{2\ell n 2}$ (2) $\frac{5}{3\ell n 2}$

(3) $\frac{5}{4\ell n 2}$ (4) $\frac{5}{6\ell n 2}$

21. A wire of resistance *R* is connected to a potential difference *V*. The drift velocity of electrons is v_d. If the wire is stretched to double its length and the same potential difference V is applied, what will be the new drift velocity?

(1) v_d (2) $\frac{v_d}{2}$

(3) $\frac{v_d}{4}$ (4) $2v_d$

22. A parallel plate capacitor with air as the dielectric is considered. Match the changes in physical quantities in **List-I** with the factor given in **List-II**.

List-I (Condition & Quantity)		**List-II (Factor)**	
(A)	Capacitance (*C*), if the distance between plates is doubled ($d \to 2d$).	(I)	Becomes 2 times
(B)	Electric Field (*E*), if the battery is disconnected and distance is doubled ($d \to 2d$).	(II)	Becomes 1/2 times
(C)	Potential Difference (*V*), if the battery is disconnected and distance is doubled ($d \to 2d$).	(III)	Remains same
(D)	Stored Energy (*U*), if the battery remains connected and a dielectric of $K = 4$ is inserted.	(IV)	Becomes 4 times

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-I, D-IV

(2) A-II, B-I, C-IV, D-III

(3) A-III, B-II, C-I, D-IV

(4) A-I, B-III, C-II, D-IV

23. A parallel plate capacitor has a capacitance $C_0 = 10\mu F$ with air between the plates. A sheet of thickness equal to half the separation between the plates $t = \frac{d}{2}$ is inserted. Which of the following materials for the sheet will result in the maximum new capacitance?

(1) A glass sheet ($K = 5$)

(2) A mica sheet ($K = 6$)

(3) A copper sheet

(4) An ebonite sheet ($K = 3$)

24. Ohm's law is obeyed by many substances, but it is not a fundamental law of nature. It fails if (where *V* is potential difference and *I* is current):

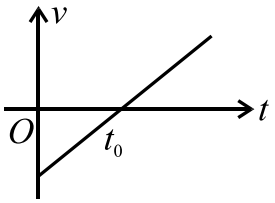
(1) *V* depends on *I* non-linearly.

(2) The relation between *V* and *I* depends on the sign of *V* for the same absolute value of *V*.

(3) The relation between *V* and *I* is non-unique.

(4) All of these.

25. Consider a particle moving along *x*-axis. If its velocity *vs* time graph is as shown in figure, then which statement can possibly be **correct**?



(1) speed of particle increases continuously.

(2) acceleration of particle changed from negative to positive at $t = t_0$.

(3) particle crosses its initial position.

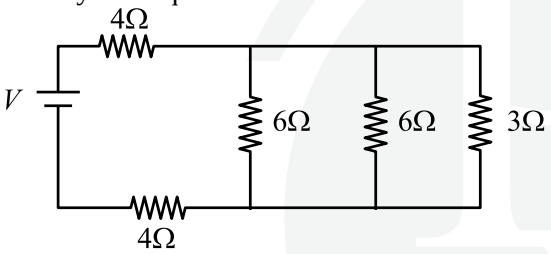
(4) its position *vs* time graph is straight line.

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26. With a significant rise in temperature, which of the following correctly describes the change in the number density of free charge carriers (n) for a metal, a semiconductor, and an insulator, respectively?
(1) Increases, decreases, and decreases
(2) Decreases, increases, and increases
(3) Remains nearly constant, increases, and increases
(4) Increases, remains constant, and remains constant

27. A capacitor of capacitance C is charged to a potential V and disconnected. It is now connected to an uncharged capacitor of capacitance $2C$. What is the percentage loss in stored energy?
(1) 33.3% (2) 50%
(3) 66.7% (4) 25%

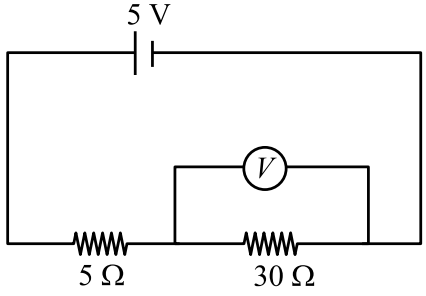
28. In the ladder resistor network shown, current through 3Ω resistor is 2 A. The emf of the ideal battery V is equal to:



(1) 20 V (2) 29 V
(3) 38 V (4) 14 V

29. If equation of motion of a particle is represented by $y = \sin \omega t + \cos \omega t$, then time period of its periodic motion is:
(1) $\frac{\pi}{\omega}$ (2) $\frac{2\pi}{\omega}$
(3) $\frac{\pi}{2\omega}$ (4) $\frac{4\pi}{\omega}$

30. As shown in the figure, the emf of the cell is 5 V and the internal resistance of the cell is zero. The reading of the ideal voltmeter is:



(1) 12 V (2) $\frac{30}{7}$ V
(3) $\frac{4}{5}$ V (4) $\frac{23}{4}$ V

31. A wire of resistance R has a temperature coefficient of resistivity α. A second wire of resistance $2R$ has a temperature coefficient 2α. These are connected in series. The effective temperature coefficient of the combination is:
(1) α (2) $\frac{3\alpha}{2}$
(3) $\frac{5\alpha}{3}$ (4) 2α

32. The value of bulk modulus of elasticity is infinite for:
(1) a perfectly rigid body.
(2) an ideal liquid.
(3) an ideal gas undergoing isochoric process.
(4) All of these.

33. The displacement of a particle executing simple harmonic motion is given by $y = B_0 + A\sin\omega t + B\cos\omega t$ (where A, B and B_0 are constant). The amplitude of its oscillation is given by: (symbols have their usual meaning)
(1) $\sqrt{A^2 + B^2}$ (2) $\sqrt{B_0^2 + (A+B)^2}$
(3) $A + B$ (4) $B_0 + \sqrt{A^2 + B^2}$

34. Given below are two statements:
Statement I: The work done by a conservative force is independent of the path taken between the initial and final position.
Statement II: Kinetic energy of body is independent of the reference frame.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

35. In meter-bridge experiment, the wire has a total length of 2 m. A resistance of 20Ω is connected in the left gap. If the balancing point is obtained at 80 cm from the left end, what is the value of the unknown resistance S in the right gap? (Assume uniform meter-bridge wire)
(1) 10Ω (2) 20Ω
(3) 30Ω (4) 40Ω

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36. Choose the **incorrect** statements among the following:

(1) Blood is more viscous than water

(2) The blood pressure in humans is greater at the feet than at the brain.

(3) The angle of contact of mercury with glass is obtuse while that of water with glass is acute

(4) A spinning cricket ball in air follows a straight trajectory.

37. A uniform wire of resistance 12Ω is bent to form a circle. Effective resistance across two points on the circumference subtending an angle of 60° at the center is:

(1) 3Ω

(2) $\frac{5}{3}\Omega$

(3) 2.5Ω

(4) 6Ω

38. If escape velocity from the earth's surface is v, then escape velocity at a height R from earth surface is: (R is the radius of earth)

(1) $\sqrt{2}v$ (2) $\frac{1}{\sqrt{2}}v$

(3) $2v$ (4) $\frac{1}{2}v$

39. The electric potential in a region is given by $V(x,y,z) = (3x^2 - y + z)$ V. The magnitude of the electric field at point $(1,2,1)$ m is:

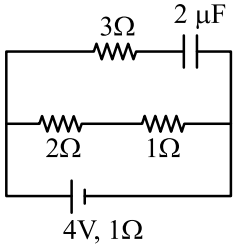
(1) 6 N/C

(2) $\sqrt{37}$ N/C

(3) $\sqrt{38}$ N/C

(4) 7 N/C

40. Calculate the potential difference across the 2 μF capacitor in the steady state condition for the circuit shown, if internal resistance of battery is 1 Ω.



(1) 5 V (2) 1 V

(3) 2 V (4) 3 V

41. Two sound waves travelling in same direction with similar amplitudes having frequencies 5 Hz and 7 Hz interfere and produce beats. What is the time interval between two consecutive maximas?

(1) 0.5 s (2) 2 s

(3) 1 s (4) 1.5 s

42. A particle moves in a circular path of radius 1 m with a constant speed of 2 m/s. The magnitude of centripetal acceleration is:

(1) 4 m/s^2 (2) 2 m/s^2

(3) 8 m/s^2 (4) 1 m/s^2

43. A stone with mass 1 kg is projected in a vertical plane at angle $\theta = 60°$ with the horizontal, with an initial speed u = 20 m/s. The angular momentum of the stone about point of projection when it is at the maximum height is: (g = 10 m/s^2)

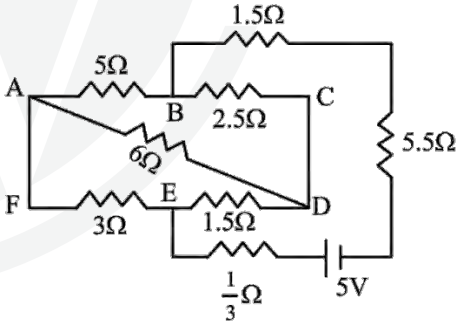
(1) 7.5 kg m^2/s

(2) 15 kg m^2/s

(3) 150 kg m^2/s

(4) 75 kg m^2/s

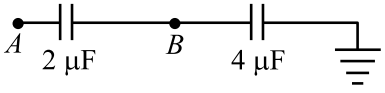
44. The current passing through the battery in the given circuit is:



(1) 2.0 A (2) 0.5 A

(3) 2.5 A (4) 1.5 A

45. Two uncharged capacitors of 2μF and 4μF are connected in series. The potential at point A is 300 V and the outer plate of 4μF is earthed as shown in figure. Then the potential at point B is:



(1) 100 V

(2) 200 V

(3) 250 V

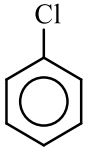
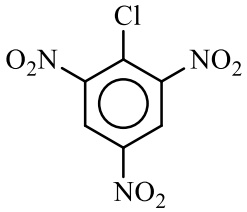
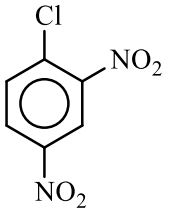
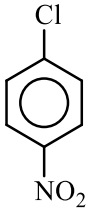
(4) 300 V

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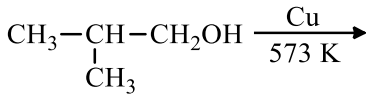
46. Given below are two statements:
Statement I: Pyrene is used as an fire extinguisher.
Statement II: Freons are used in refrigeration and air conditioning.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

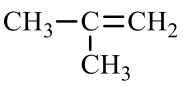
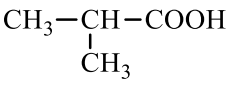
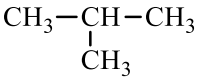
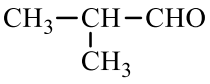
47. For the reaction A $\rightarrow$ B, if the initial concentration of A is 0.1 M and rate constant of reaction is 0.4 M s^{-1}, then time of completion of the reaction is:
(1) infinite
(2) 0.75 s
(3) 1.5 s
(4) 0.25 s

48. How many moles of $CaSO_4$ will contain 8 g of oxygen atoms?
(1) 0.25 (2) 0.125
(3) 0.165 (4) 0.375

49. Which of the following compound undergo ArS_N reaction most easily?
(1) 
(2) 
(3) 
(4) 

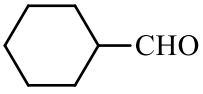
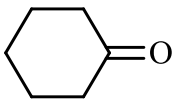
50. Major product of the following reaction is:



(1) 
(2) 
(3) 
(4) 

51. Which of the following orbitals has maximum energy?
(1) n = 4, ℓ = 1, m = +1
(2) n = 5, ℓ = 0, m = 0
(3) n = 3, ℓ = 2, m = –1
(4) n = 4, ℓ = 0, m = 0

52. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: A catalyst increases the Gibb's energy of the reaction.
Reason R: A catalyst does not change the equilibrium constant of a reaction.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

53. Which among the following compounds give 3°-alcohol on treatment with CH_3MgCl followed by hydrolysis?
(1) 
(2) 
(3) HCHO
(4) CH_3CHO

[Contd...

54. Aqueous solution of which of the following salts turn red litmus paper blue?
(1) NH_4Cl
(2) NaCl
(3) KCN
(4) K_2SO_4

55. The rate constant of a first order reaction with takes 4.606 seconds for 90% completion is:
(1) 0.25 s^{-1}
(2) 1.5 s^{-1}
(3) 0.8 s^{-1}
(4) 0.5 s^{-1}

56. Most acidic compound among the following is:
(1) o-nitrophenol
(2) p-cresol
(3) p-nitrophenol
(4) m-cresol

57. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Oxidation state of Cr in CrO_5 is +6.
Reason R: CrO_5 contain two peroxide bond.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

58. The reaction, $N_2O_5(g) \rightarrow 2NO_2(g) + \frac{1}{2}O_2(g)$, is an example of:
(1) zero order reaction
(2) second order reaction
(3) first order reaction
(4) pseudo first order reaction

59. The **correct** statements regarding aspirin are:
(A) It can be obtain by acetylation of salicylic acid.
(B) It possesses analgesic properties.
(C) It is an isomer of methyl salicylate.
(1) A and B only
(2) B and C only
(3) A and C only
(4) A, B and C

60. Consider the following equilibrium reactions with their equilibrium constants

$X(g) \overset{K_1}{\rightleftharpoons} 2Y(g)$

$Y(g) \overset{K_2}{\rightleftharpoons} 4Z(g)$

The equilibrium constant for the reaction $4Z(g) \rightleftharpoons \frac{1}{2}X(g)$, will be:

(1) $\frac{1}{K_1\sqrt{K_2}}$

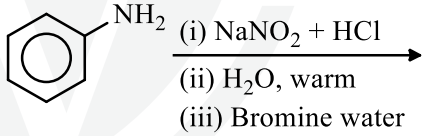
(2) $\frac{1}{K_2\sqrt{K_1}}$

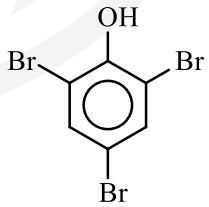
(3) $\frac{1}{\sqrt{K_1K_2}}$

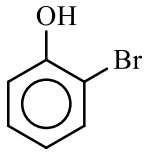
(4) $\frac{2}{K_1K_2}$

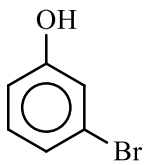
61. For a chemical reaction with rise in temperature by 10°C, the rate constant is nearly:
(1) tripled
(2) doubled
(3) quadrupled
(4) remain same

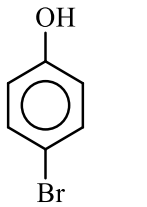
62. The major product (X) in the following reaction sequence is:



(1) 

(2) 

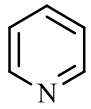
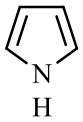
(3) 

(4) 

[Contd...

63. The commercial alcohol is made unfit for drinking (denaturation) by mixing in it:

(A) $CuSO_4$ (B) $MgSO_4$

(C)  (D) 

(1) A and D (2) B and C

(3) A and C (4) B and D

64. Achiral molecule among the following is:

(1) 1-Chloropropane

(2) Butan-2-ol

(3) 2-Hydroxybutanal

(4) 2-Bromopropanoic acid.

65. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:

Assertion A: Ortho and para nitrophenol can be separated by steam distillation.

Reason R: o-Nitrophenol is steam volatile due to intramolecular H-bonding while p-nitrophenol is less volatile due to intermolecular H-bonding.

In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

66. Which of the following discriminate between reversible and irreversible expansion of an ideal gas?

(1) ΔU and W (2) W and ΔS

(3) ΔU and ΔS (4) W and ΔU

67. Given below are two statements:

Statement I: Diastereomers are optically active stereoisomers.

Statement II: Racemic mixture may be optically active.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

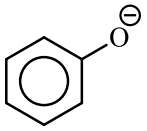
(1) Both Statement I and Statement II are correct.

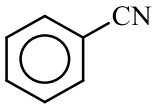
(2) Statement I is correct but Statement II is incorrect.

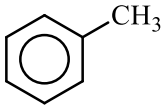
(3) Statement I is incorrect but Statement II is correct.

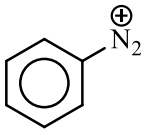
(4) Both Statement I and Statement II are incorrect.

68. Which of the following is more reactive towards electrophilic aromatic substitution than phenol?

(1) 

(2) 

(3) 

(4) 

69. IUPAC group prefix used for ethers is:

(1) keto

(2) alkoxy

(3) alkoxy carbonyl

(4) carbonyl

70. Given below are two statements:

Statement I: Molecularity and order of an elementary reaction are same.

Statement II: Molecularity is defined for both elementary as well as complex reactions.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are incorrect.

71. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:

Assertion A: The C–O–C bond angle in CH_3OCH_3 is slightly less than the tetrahedral angle.

Reason R: There is repulsive interaction between the two lone pair of oxygen atoms.

In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

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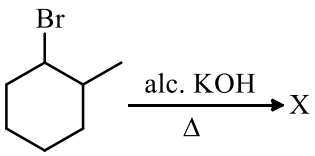
72. Maximum possible structural isomers of alkane corresponding to the molecular formula C_5H_{12} is:
(1) 2
(2) 3
(3) 4
(4) 5

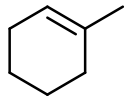
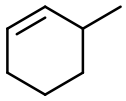
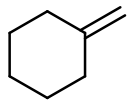
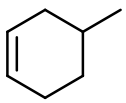
73. The **correct** statements among the following are:
(A) Dry acetone is used as solvent in Finkelstein reaction.
(B) Cyanide behave as an ambidentate nucleophile.
(C) 1,2-Dichlorobenzene has more melting point than 1,4-dichlorobenzene.
(1) A and B only
(2) A and C only
(3) B and C only
(4) A, B and C

74. Williamson synthesis is used for the preparation of:
(1) Alcohols
(2) Ethers
(3) Aldehydes
(4) Carboxylic acids

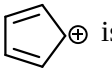
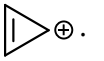
75. Given below are two statements:
Statement I: An ideal black body, emits and absorb radiations of all frequencies uniformly.
Statement II: For an ideal black body, the intensity of radiation depends on its temperature and surface area.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

76. Major product (X) of the following reaction is:



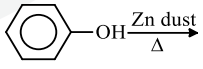
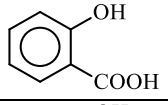
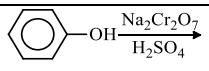
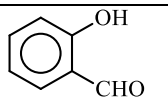
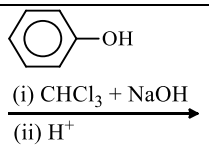
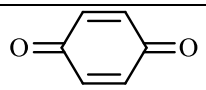
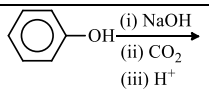
(1) 
(2) 
(3) 
(4) 

77. Major products of the following reaction are:
$CH_3-O-CH_2CH_3 \xrightarrow{\text{Conc. HI}}$
(1) $CH_3I + CH_3CH_2OH$
(2) $CH_3OH + CH_3CH_2I$
(3) $CH_3OH + CH_2=CH_2$
(4) $CH_3I + CH_2=CH_2$

78. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A:  is more stable than  .
Reason R:  has more number of π-electrons than  .
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

79. For the same alkyl group, the boiling points of alkyl halides decreases in the order:
(1) RI > RF > RBr > RCl
(2) RI > RBr > RF > RCl
(3) RI > RBr > RCl > RF
(4) RF > RI > RBr > RCl

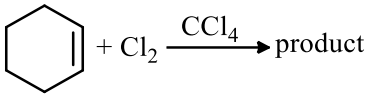
80. Match **List-I** with **List-II**.

	List-I (Reaction)		**List-II (Product)**
(A)		(I)	
(B)		(II)	
(C)		(III)	
(D)		(IV)	

Choose the **correct** answer from the options given below:
(1) A-III; B-IV; C-II; D-I
(2) A-IV; B-III; C-II; D-I
(3) A-IV; B-I; C-II; D-III
(4) A-III; B-I; C-IV; D-II

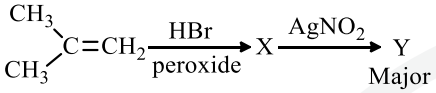
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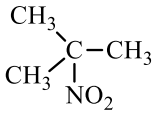
81. The **correct** option for the given reaction is:



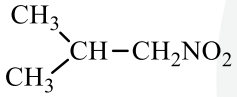
(1) Electrophilic addition reaction with carbocation intermediate
(2) Electrophilic addition reaction with cyclic halonium ion intermediate
(3) Free radical addition reaction with free radical intermediate
(4) rearrangement reaction with carbocation intermediate

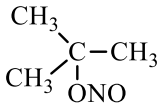
82. The product (Y) formed in the following reaction sequence is:



(1) 

(2) 

(3) 

(4) 

83. A better reagent for oxidation of primary alcohols to aldehydes in good yield is:
(1) alkaline $KMnO_4$
(2) PCC
(3) acidified $K_2Cr_2O_7$
(4) $NaBH_4$

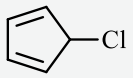
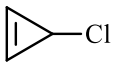
84. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Boiling point of n-butane is more than that of propane.
Reason R: There is more van der Waal forces among n-butane molecules due to more surface area than propane.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

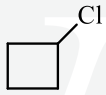
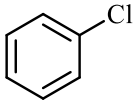
85. How many enantiomeric pair can be obtain on monochlorination of isopentane?
(1) 1
(2) 2
(3) 3
(4) 4

86. Nowadays, large quantity of ethanol are obtained by:
(1) hydration of ethene
(2) fermentation of sugars
(3) catalytic reduction of ethanal
(4) hydroboration oxidation of ethene

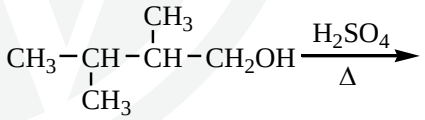
87. Which of the following is **not** a colligative property?
(1) Osmotic pressure
(2) Depression in freezing point
(3) Vapour pressure
(4) Relative lowering of vapour pressure

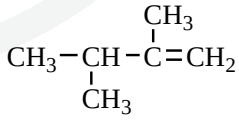
88. Most reactive compound towards S_N1 reaction among the following is:

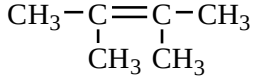
(1)  (2) 

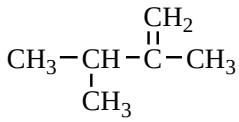
(3)  (4) 

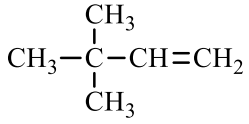
89. Major product of the following reaction is:



(1) 

(2) 

(3) 

(4) 

90. The number of faraday required for the conversion of 0.5 mol I_2 to IO_4^- is:
(1) 14 F (2) 7 F
(3) 3.5 F (4) 9 F

[Contd...

91. Identify the **correct** statements regarding the Hershey-Chase experiment.

A. Viruses grown in radioactive phosphorus contained radioactive DNA.
B. Radioactivity was detected in the supernatant for the viruses grown on radioactive sulfur after centrifugation.
C. Bacteria infected with viruses that had radioactive DNA were not radioactive.
D. The experiment concluded that protein is the genetic material that is passed from virus to bacteria.
E. Viral coats were removed from the bacteria by agitating them in a blender.

Choose the *most appropriate* answer from the options given below.

(1) A, B and E only
(2) C and D only
(3) A, D and E only
(4) B, C and D only

92. The protists responsible for formation of diatomaceous earth;

(1) also cause red tides in sea.
(2) are classified as chrysophytes.
(3) do not possess photosynthetic machinery.
(4) live as parasites or predators.

93. In flowering plants, endosperm development precedes embryo development because:

(1) it forms the protective seed coat around the embryo.
(2) it is an adaptation to provide assured nutrition to the developing embryo.
(3) it helps in the dispersal of seeds to new habitats.
(4) it is necessary for the development of the fruit wall or pericarp.

94. Select the **correctly** matched pair.

(1) Monocot root - Pith is small and inconspicuous
(2) Dicot root - Usually has more than six xylem bundles
(3) Monocot root - Undergoes secondary growth
(4) Dicot root - Stele consists of pericycle, vascular bundles and pith

95. Choose the **incorrect** statement about *E. coli*.

(1) It has 4.6×10^6 bp in its DNA.
(2) It was the first organism in which semi-conservative replication of DNA was proved.
(3) It divides in approximately 20 seconds.
(4) In it, transcription and translation can be coupled.

96. Identify the **incorrectly** matched pair.

(1) *Volvox* - Female gamete is static and male gamete is motile
(2) *Pinus* - Both male and female gametes are non-motile
(3) *Funaria* - Male gamete is flagellated and egg is non-motile
(4) *Selaginella* - Both male and female gametes are motile

97. Which of the following are observed post-fertilisation in flowering plants?

A. The zygote develops into the embryo.
B. The central cell of embryo sac develops into the endosperm.
C. The ovules mature into seeds.
D. The ovary develops into the fruit.
E. The synergids and antipodals form the seed coat.

Choose the *most appropriate* answer from the options given below.

(1) A, B, C and D only
(2) C, D and E only
(3) A, D and E only
(4) A, B and C only

98. In a bacterial cell, the 'ρ' factor was not produced due to a mutation in its gene. It will directly impact;

(1) replication of DNA.
(2) termination of transcription in it.
(3) initiation of transcription in it.
(4) joining of discontinuously synthesised DNA fragments.

99. Match **List-I** with **List-II**.

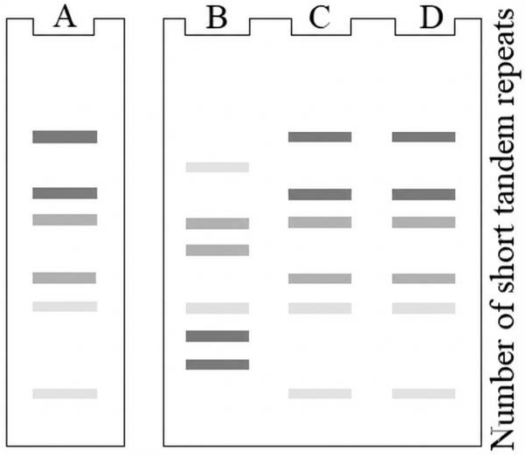
List-I		List-II	
(A)	Leptotene	(I)	Meiotic spindle is assembled
(B)	Zygotene	(II)	Can last for months or years in oocytes of some vertebrates
(C)	Diplotene	(III)	Chromosomes become gradually visible under the light microscope
(D)	Diakinesis	(IV)	Tetrad of chromosomes formed

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-IV, C-II, D-I
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-I, C-IV, D-II
(4) A-IV, B-I, C-II, D-III

100. For a typical angiosperm, how many meiotic and mitotic divisions in nucleus are required to form 100 mature female gametophytes from megaspore mother cells?
(1) 100 meiotic and 100 mitotic divisions
(2) 100 meiotic and 200 mitotic divisions
(3) 100 meiotic and 300 mitotic divisions
(4) 400 meiotic and 400 mitotic divisions

101. Analyse the DNA fingerprinting result shown in the gel image below. 'A' is the sample of DNA from crime site and B, C and D are suspects. Identify the true statement about it.



(1) C and D are identical twins.
(2) B is the criminal.
(3) Either C or D can be the criminal, but not both.
(4) The level of DNA polymorphism is variable in A, C and D.

102. Identify the ranks of the following taxonomic categories and arrange them in their correct hierarchical sequence from the highest to the lowest.
A. Pteropsida
B. Felidae
C. Fungi
D. *Brassica*
E. Primata
Choose the **correct** sequence from the options below.
(1) C → E → A → B → D
(2) C → A → E → B → D
(3) D → B → E → A → C
(4) C → A → B → E → D

103. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In some plant species, fruits develop without fertilisation.
Reason (R): Application of growth hormones can induce the development of seedless fruits.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

104. Match **List-I** with **List-II** regarding the salient features of the Human Genome Project.

List-I		List-II	
(A)	Human genome content	(I)	Less than 2%
(B)	Total number of genes	(II)	3164.7 million bp
(C)	Percentage of genome coding for proteins	(III)	99.9%
(D)	Percentage of identical nucleotide bases in all humans	(IV)	Approximately 30,000

Choose the *most appropriate* answer from the options given below.
(1) A-II, B-III, C-I, D-IV
(2) A-II, B-IV, C-I, D-III
(3) A-I, B-II, C-IV, D-III
(4) A-III, B-I, C-II, D-IV

105. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In a hypogynous flower, the ovary is said to be superior.
Reason (R): In a hypogynous flower, the gynoecium occupies the highest position while other floral parts are situated below it.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

106. Given below are two statements:
Statement I: Insects are the most dominant abiotic agents of pollination.
Statement II: Majority of plants use biotic agents for pollination.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

[Contd...

107. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): DNA is chemically less reactive and structurally more stable as compared to RNA.
Reason (R): The presence of uracil at the place of thymine confers additional stability to DNA.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

108. Read the following statements.
A. Theodore Schwann was a German botanist.
B. Rudolf Virchow had no role in formulation of cell theory as understood today.
C. Robert Brown was the first to explain that cells divide and new cells are formed from pre-existing cells.
D. Matthias Schleiden was a German zoologist.
E. Both prokaryotic and eukaryotic cells have a semi-fluid matrix called cytoplasm.
How many of the above statements are **correct**?
(1) One (2) Two
(3) Three (4) Four

109. All the following are true about the genetic code, **except** that it;
(1) is degenerate.
(2) is nearly universal.
(3) has punctuations.
(4) is a triplet.

110. Identify the **incorrect** statement about bryophytes.
(1) They are called amphibians of the plant kingdom.
(2) The main plant body is a diploid sporophyte.
(3) They lack true roots, stem, or leaves.
(4) The sporophyte derives nourishment from the gametophyte.

111. The sporopollenin;
(1) is present in the intine of pollen grain.
(2) is the most resistant organic material known.
(3) is crucial for the storage of seeds.
(4) can be degraded enzymatically.

112. A typical nucleosome contains ____ bp of DNA helix.
(1) 300 (2) 500
(3) 400 (4) 200

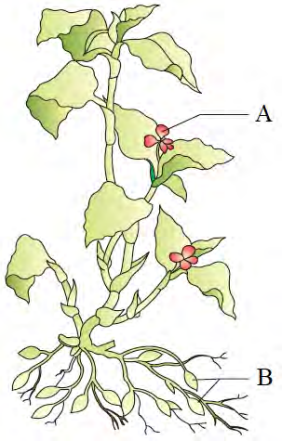
113. In a dorsiventral leaf, the;
(1) tissue between the upper and lower epidermis is parenchymatous.
(2) adaxial epidermis lacks cuticle.
(3) abaxial epidermis lacks stomata.
(4) venation is usually parallel.

114. Identify the **odd one out** w.r.t ploidy in the given structures of angiosperms.
(1) Antipodal cell
(2) Microspore
(3) Megaspore mother cell
(4) Egg cell

115. Given below are two statements:
Statement I: QB bacteriophage has RNA as the genetic material.
Statement II: *Caenorhabditis elegans* is a free living non-pathogenic nematode.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

116. Which of the following species is not associated with the genus *Solanum*?
(1) *nigrum* (2) *tuberosum*
(3) *melongena* (4) *aestivum*

117. Choose the **correct** statement for the plant shown below.



(1) It produces both unisexual and bisexual flowers.
(2) In it, autogamy cannot occur.
(3) Geitonogamy brings about genetic variations in its progeny.
(4) Xenogamy can occur in flower A but not in B.

[Contd...

118. Which of the following are true regarding post-transcriptional modifications in eukaryotes?

A. Splicing removes exons and joins introns.
B. Capping takes place at 5'-end of hnRNA.
C. Tailing takes place at 3'-end of hnRNA.
D. Capping involves addition of methyl guanosine diphosphate.
E. Tailing involves addition of 2000-3000 adenine residues.

Choose the *most appropriate* answer from the options given below.

(1) A and E
(2) B and C
(3) C and D
(4) A and C

119. The morphology of chromosomes is most easily studied at:

(1) Prophase.
(2) Metaphase.
(3) Anaphase.
(4) Telophase.

120. The ears of a corn cob are adapted for;

(1) wind-pollination.
(2) water-pollination.
(3) dispersal of pollen grains.
(4) insect-pollination.

121. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Frederick Griffith	(I)	Proposed the double helix model for DNA structure
(B)	Watson and Crick	(II)	Observed transformation in pneumococcus
(C)	Frederick Sanger	(III)	Performed biochemical characterisation of the transforming principle
(D)	Avery, MacLeod and McCarty	(IV)	Developed methodology for sequencing of DNA

Choose the *most appropriate* answer from the options given below.

(1) A-II, B-I, C-IV, D-III
(2) A-II, B-IV, C-I, D-III
(3) A-I, B-II, C-IV, D-III
(4) A-III, B-I, C-II, D-IV

122. Given below are two statements:

Statement I: The endomembrane system includes the Golgi complex and vacuoles.

Statement II: The functions of mitochondria and chloroplasts are coordinated with the endomembrane system.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

123. Water is the pollinating agent in;

A. water lily.
B. *Hydrilla*.
C. *Zostera*.
D. water hyacinth.
E. *Vallisneria*.

Choose the *most appropriate* answer from the options given below.

(1) C only
(2) A and B only
(3) D and E only
(4) B, C and E only

124. The process that requires transfer of genetic information from a polymer of nucleotides to synthesise a polymer of amino acids is called:

(1) transcription.
(2) replication.
(3) translation.
(4) transformation.

125. When the placenta is axial and the ovules are attached to it in a multilocular ovary, the placentation is said to be;

(1) marginal.
(2) parietal.
(3) basal.
(4) axile.

126. The plant *Amorphophallus*;

(1) provides rewards for pollinators in the form of safe place for laying eggs.
(2) produces the smallest flower.
(3) is mainly pollinated by wind or water.
(4) is parasitic.

[Contd...

127. The sensitivity of DNA fingerprinting has been increased by use of the technique;
(1) reverse transcription.
(2) electrophoresis.
(3) polymerase chain reaction.
(4) southern blotting.

128. Which of the following statements regarding fungi are **correct**?
A. Phycomycetes have coenocytic and aseptate mycelium.
B. *Aspergillus* is a member of Basidiomycetes.
C. In Ascomycetes, sexual spores are produced exogenously.
D. Deuteromycetes reproduce only by asexual spores known as conidia.
E. Yeast is a multicellular fungus belonging to Ascomycetes.
Choose the *most appropriate* answer from the options given below.
(1) A and D only
(2) B and C only
(3) C, D and E only
(4) A, B and E only

129. Sexual reproduction enables:
(1) individual organisms to live forever.
(2) the creation of new variants so that survival advantage is enhanced.
(3) increased vulnerability to extinction.
(4) the conservation of genome amongst generations.

130. Identify the correct statements from the following.
A. A nucleotide has a nitrogenous base, a pentose sugar, and a phosphate group.
B. Adenine is a pyrimidine, common to both DNA and RNA.
C. A nitrogenous base is linked to the pentose sugar through a N-glycosidic linkage to form a nucleoside.
D. A phosphate group is linked to the 5'-OH of a nucleoside through a phosphoester linkage.
E. Two consecutive nucleotides are linked by a hydrogen bond.
Choose the *most appropriate* answer from the options given below.
(1) A, B and C only
(2) B and E only
(3) A, D and E only
(4) A, C and D only

131. The ovule is attached to the placenta by a stalk called the:
(1) hilum.
(2) funicle.
(3) micropyle.
(4) petiole.

132. Euchromatin is;
(1) the densely packed chromatin.
(2) the chromatin that stains dark.
(3) transcriptionally inactive chromatin.
(4) transcriptionally active chromatin.

133. Match **List-I** with **List-II**.

List-I		**List-II**	
(A)	True fruit	(I)	Strawberry
(B)	False fruit	(II)	Mango
(C)	Apomixis	(III)	Orange
(D)	Polyembryony	(IV)	Some species of *Asteraceae*

(1) A-II, B-I, C-IV, D-III
(2) A-III, B-IV, C-I, D-II
(3) A-I, B-II, C-III, D-IV
(4) A-IV, B-III, C-II, D-I

134. Given below are two statements:
Statement I: A transcription unit in DNA is defined primarily by a promoter, the structural gene, and a terminator.
Statement II: The promoter is located towards the 3'-end (downstream) of the structural gene.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

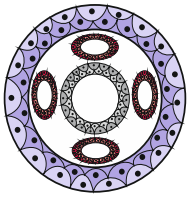
135. The seed excavated from the Arctic Tundra;
(1) was 100,000 years old.
(2) germinated and flowered after an estimated record of 10,000 years of dormancy.
(3) was that of *Phoenix dactylifera*.
(4) was discovered near dead sea.

136. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The secondary immune response is faster and more intense than the primary immune response.
Reason R: This is due to the presence of memory B and T cells generated during the first encounter which recognize the pathogen quickly.
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

137. Read the given statements about enzymatic activity.
A. A low Km indicates high affinity of an enzyme for its substrate.
B. Zinc is co-factor of enzyme carboxypeptidase.
C. Enzymes bring down the activation energy barrier making the transition of 'S' to 'P' more difficult.
D. Formation of glucose to pyruvic acid is a metabolic pathway consisting of ten reactions.
How many of the above statements are **correct**?
(1) One (2) Two
(3) Three (4) Four

138. Given below are two statements.
Statement I: ZIFT involves the transfer of a zygote or early embryo with up to 8 blastomeres into the fallopian tube.
Statement II: IUI involves the transfer of embryos with more than 8 blastomeres into the uterus.
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

139. Identify the type of coelom represented in the following diagram.



(1) Acoelomate
(2) Eucoelomate
(3) Pseudocoelomate
(4) Schizocoelomate

140. Match **List-I** with **List-II**.

List-I		List-II	
(A)	PCT	(I)	Conditional reabsorption of Na^+ and water
(B)	Descending Limb of Loop of Henle	(II)	Reabsorption of 70-80% electrolytes
(C)	DCT	(III)	Permeable to water, impermeable to electrolytes
(D)	Ascending Limb of Loop of Henle	(IV)	Impermeable to water, permeable to electrolytes

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-I, D-III
(2) A-II, B-III, C-I, D-IV
(3) A-I, B-III, C-II, D-IV
(4) A-III, B-II, C-I, D-IV

141. The **best** description of contact inhibition is:
(1) The ability of cancer cells to detach and spread to other organs.
(2) The property of normal cells to cease growth when they touch other cells.
(3) The ability of tumors to starve normal cells of nutrients.
(4) The induction of angiogenesis by tumor cells.

142. The sugar present in RNA has the chemical formula:
(1) $C_6H_{12}O_6$ (2) $C_5H_{10}O_4$
(3) $C_5H_{10}O_5$ (4) $C_4H_8O_4$

143. Which of the following contraceptive methods works by releasing copper ions to suppress sperm motility and fertilizing capacity?
(1) LNG-20
(2) Multiload 375
(3) Saheli
(4) Progestasert

144. A lesion in a specific part of the brain leads to unaffected voluntary movements but causes severe impairment in emotional responses and sexual behaviour, with intact respiration and cardiovascular reflexes. The **most** likely affected region is:
(1) Cerebral aqueduct
(2) Limbic system
(3) Medulla oblongata
(4) Pons

[Contd...

145. Given below are two statements.
Statement I: Spermatogenesis starts at the age of puberty due to significant increase in the secretion of GnRH.
Statement II: Oogenesis is initiated during the embryonic development stage when a couple of million gamete mother cells (oogonia) are formed within each foetal ovary.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

146. Identify the **correct** sequence of stages in the life cycle of *Plasmodium*:
A. Sporozoites released from mosquito salivary glands.
B. Parasite reproduces asexually in liver cells.
C. Sexual stages (gametocytes) develop in human RBCs.
D. Female mosquito takes up gametocytes with blood meal.
E. Rupture of RBCs and release of haemozoin.
(1) A→ E → B → C → D
(2) A→ B → E → C → D
(3) A→ B → C → E → D
(4) B → A→ E → D → C

147. Which of the following is **not** a second messenger w.r.t membrane bound receptor hormones?
(1) Cyclic AMP (2) IP_3
(3) Calcium ions (4) Trypsin

148. Identify the **mismatched** pair regarding secondary metabolites:
(1) Pigments: Carotenoids, Anthocyanins
(2) Alkaloids: Morphine, Codeine
(3) Toxins: Abrin, Ricin
(4) Lectins: Vinblastin, Curcumin

149. Identify the **incorrect** statement.
(1) Abortions could happen spontaneously too.
(2) Infertility is defined as the inability to produce a viable offspring and is always due to abnormalities/defects in the female partner.
(3) Complete lactation up to 6 months post parturition could help as a natural method of contraception.
(4) Creating awareness about sex related aspects is an effective method to improve reproductive health of the people.

150. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Every 100 ml of oxygenated blood can deliver around 5 ml of O_2 to the tissues under normal physiological conditions.
Reason R: Binding of oxygen with haemoglobin is primarily related to partial pressure of O_2, and partial pressure of CO_2 does not interfere with this binding.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

151. Identify the lipid that contains a phosphorus and a phosphorylated organic compound in it, found in cell membranes.
(1) Palmitic acid (2) Glycerol
(3) Lecithin (4) Arachidonic acid

152. During the replication of retrovirus, the viral RNA genome is converted into viral DNA by the enzyme:
(1) DNA polymerase
(2) Protease
(3) Reverse transcriptase
(4) Integrase

153. Find the **correct** statements about the cardiac cycle.
A. The duration of a cardiac cycle is 0.8 seconds.
B. The stroke volume is approximately 70 mL.
C. The lub sound is associated with the closure of the semilunar valves.
D. The P-wave in an ECG represents the depolarization of the atria.
(1) A, B and C (2) A, B and D
(3) B, C and D (4) A, C and D

154. The statutory ban on amniocentesis is primarily to:
(1) Prevent genetic disorders like Down syndrome.
(2) Check female foeticide.
(3) Detect survivability of the foetus.
(4) Improve reproductive health.

155. According to the sliding filament theory, during muscle contraction, which of the following zones/bands retains its length?
(1) I-band (2) A-band
(3) H-zone (4) Sarcomere

[Contd...

156. Identify the **incorrect** statement regarding protein structure.
(1) The primary structure describes the positional information of amino acids.
(2) The alpha-helix is a type of secondary structure where the chain is coiled left-handedly.
(3) Tertiary structure is necessary for the biological activity of proteins.
(4) Haemoglobin exhibits quaternary structure.

157. Which drug binds to specific receptors present in the CNS and gastrointestinal tract, often functioning as a depressant and slowing down body functions?
(1) Heroin (2) Cocaine
(3) Marijuana (4) *Datura*

158. Protonephridia (flame cells) are the excretory structures in:
(1) Cockroaches (2) Platyhelminthes
(3) Prawns (4) Earthworms

159. Saheli, a contraceptive pill developed in India, is characterised by which of the following features?
(1) It is a progesterone-only pill taken daily
(2) It is a non-steroidal oral contraceptive taken once a week
(3) It inhibits ovulation by increasing LH surge
(4) It causes permanent infertility in females

160. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Cellulose is a homopolymer.
Reason R: The glucose and fructose units are present in cellulose.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

161. Which diagnostic test confirms typhoid fever, and who was the famous carrier of this disease?
(1) ELISA; Mary Mallon
(2) Widal; Mary Mallon
(3) Biopsy; Edward Jenner
(4) Widal; Louis Pasteur

162. Which of the following **correctly** describes the central nervous system in chordates and non-chordates?

	Chordates (CNS)	**Non-chordates (CNS)**
(1)	Dorsal, hollow, single	Ventral, solid, paired
(2)	Ventral, solid, paired	Dorsal, hollow, single
(3)	Dorsal, solid, paired	Ventral, hollow, single
(4)	Ventral, hollow, paired	Dorsal, solid, single

163. Which of the following STIs is **not** completely curable even if detected early?
(1) Syphilis (2) Gonorrhoea
(3) Genital Herpes (4) Chlamydiasis

164. Match **List-I** with **List-II**.

List-I		**List-II**	
(A)	Monosaccharide	(I)	Heteropolymer of nucleotides
(B)	Proteins	(II)	$CH_3–(CH_2)_{14}–COOH$
(C)	Fatty acid	(III)	$C_6H_{12}O_6$
(D)	Nucleic Acids	(IV)	Peptide bond

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-II, D-I
(2) A-III, B-II, C-IV, D-I
(3) A-IV, B-III, C-II, D-I
(4) A-II, B-IV, C-III, D-I

165. The four small glands located on the back side of the thyroid gland secrete a hormone that:
(1) Lowers blood calcium levels
(2) Increases blood calcium levels
(3) Regulates basal metabolic rate
(4) Stimulates erythropoiesis

166. The antibody primarily present in colostrum is:
(1) IgG (2) IgM
(3) IgD (4) IgA

167. Which of the following are contributing factors to population explosion in India?
A. Rapid decline in death rate.
B. Rapid decline in maternal mortality rate.
C. Decrease in the number of people in reproducible age.
D. Rapid decline in infant mortality rate.
(1) A and B only (2) A, B and D only
(3) C and D only (4) A, B, C and D

168. Renin released from JG cell converts:
(1) Angiotensin I to Angiotensin II
(2) Angiotensinogen to Angiotensin I
(3) Angiotensin II to Aldosterone
(4) Angiotensinogen to Angiotensin II directly

169. Which of the following **correctly** distinguishes essential and non-essential amino acids?
(1) Essential amino acids can be synthesized by the body; non-essential amino acids cannot.
(2) Essential amino acids cannot be synthesized by the body and must be obtained from diet; non-essential amino acids can be synthesized by the body.
(3) Both essential and non-essential amino acids cannot be synthesized by the body.
(4) Both essential and non-essential amino acids must be obtained entirely from the diet.

[Contd...

170. Select the **correctly** matched pair of Disease and Vector:
(1) Malaria - Female *Culex*
(2) Filariasis - Female *Anopheles*
(3) Dengue - Female *Aedes*
(4) Amoebiasis - Female *Aedes*

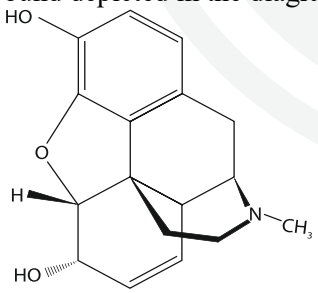
171. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Fibrous Joint	(I)	Between Atlas and Axis
(B)	Cartilaginous Joint	(II)	Between Humerus and Pectoral Girdle
(C)	Pivot Joint	(III)	Skull sutures
(D)	Ball and Socket	(IV)	Between adjacent vertebrae

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-III, B-I, C-IV, D-II
(3) A-IV, B-III, C-I, D-II
(4) A-III, B-IV, C-II, D-I

172. Which of the following statement is **true**?
(1) Surgical methods of contraception prevent gamete formation.
(2) All sexually transmitted diseases are caused by bacteria
(3) Oral contraceptive pills are very effective with lesser side effects and are well accepted by the females.
(4) In embryo transfer (ET) techniques, embryos are sometimes discarded outside the body.

173. Identify the product obtained upon acetylation of the compound depicted in the diagram.



(1) Marijuana (2) Hashish
(2) Charas (4) Heroin

174. Allergy is due to the release of chemicals like histamine and serotonin from:
(1) Lymphocytes (2) Mast cells
(3) RBCs (4) Platelets

175. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Tidal Volume	(I)	2500 - 3000 mL
(B)	IRV	(II)	1100 - 1200 mL
(C)	ERV	(III)	500 mL
(D)	Residual Volume	(IV)	1000 - 1100 mL

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-IV, D-II
(2) A-III, B-I, C-II, D-IV
(3) A-I, B-III, C-IV, D-II
(4) A-III, B-IV, C-I, D-II

176. Which of the following is a reusable barrier method for females?
(1) Female Condom (2) Diaphragms
(3) Progestasert (4) Implants

177. Consider the following statements.
A. ABO grouping is based on the presence or absence of surface antigens on the RBCs.
B. Plasma contains antibodies against the antigens absent on the RBCs.
C. O- blood group is the universal donor.
D. *Erythroblastosis foetalis* occurs when the mother is Rh^+ and the foetus is Rh-.
Which statements are **correct**?
(1) A, B and C only (2) A, B and D only
(3) B and D only (4) A, B, C and D

178. In competitive inhibition, the inhibitor molecule bears a close structural resemblance to which of the following?
(1) The product
(2) The enzyme's active site
(3) The co-factor
(4) The substrate

179. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Mucosa-associated lymphoid tissue (MALT) constitutes about 50 percent of the lymphoid tissue in the human body.
Reason R: Spleen and thymus are part of MALT.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

180. During the resting and depolarisation stages of a neuron, which of the following **correctly** describes the membrane permeability to ions respectively?

	Resting stage permeability	Depolarisation stage permeability
(1)	High Na^+ permeability	High K^+ permeability
(2)	High K^+ permeability	High Na^+ permeability
(3)	Equal Na^+ and K^+ permeability	Decreased Na^+ permeability
(4)	Impermeable to ions	Permeable only to Cl^-

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